

A certified upper bound for the real Grothendieck constant

Abstract

An explicit Gaussian rounding scheme gives

$$K_G^{\mathbb{R}} \leq \frac{\pi}{2(0.8818276493988)} < 1.781296297373.$$

The scheme uses a degree-nine polynomial threshold and a degree-eighty-one tensor preprocessing polynomial. The proof combines a coefficient inequality valid for every finite real matrix, interval enclosures of all 63252 Gaussian moments required through degree 501, and a bound on the fourth angular derivative on the entire unit circle. We give the Gaussian identities, quadrature remainders, boundary convergence argument, and exact rational comparisons underlying the certificate. All polynomial coefficients and the finite data needed to reproduce the calculation accompany the proof.

1 Statement and normalization

For a finite real matrix $M = (M_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}$, put

$$S(M) = \max_{\varepsilon_i, \delta_j \in \{-1, 1\}} \left| \sum_{i,j} M_{ij} \varepsilon_i \delta_j \right|,$$
$$V(M) = \max_{\|u_i\|=\|v_j\|=1} \left| \sum_{i,j} M_{ij} \langle u_i, v_j \rangle \right|.$$

The vectors are real. Their span has dimension at most $m+n$, so the latter maximum can be taken in $\mathbb{R}^{m+n}$ and is attained. The real Grothendieck constant is the least K for which $V(M) \leq KS(M)$ for every such matrix.

Theorem 1. *For every pair of positive integers m, n and every $M \in \mathbb{R}^{m \times n}$,*

$$V(M) \leq \frac{\pi}{2(0.8818276493988)} S(M).$$

In particular, $K_G^{\mathbb{R}} < 1.781296297373 < 1.78130$.

The normalization throughout is $H = (\pi/2)\mathbb{E}[fg]$ for real Gaussian rounding. Gaussian rounding and tensor preprocessing originate in work of Krivine [1]; multivariate rounding was developed further by Braverman, Makarychev, Makarychev, and Naor [2]. The combination of coordinate-dependent preprocessing, Hermite coefficients, a coefficient margin, and a Sobolev estimate follows methods used by Saha et al. [3]. The argument below proves the needed statements for the explicit polynomials used here; it does not use an upper-bound theorem from those papers as a premise.

For comparison, the numerical inequalities in Proposition 6.3 of [3], version 2, imply the upper bound

$$\frac{\pi}{2 \left(0.881573822049 - 0.000011328860 - 14.443/\sqrt{10 \cdot 251^5} \right)} = 1.78184132423347648 \dots \quad (1)$$

Li et al. [4], version 3, separately report 1.7813319810625639 as an internally tested upper bound, expressly stating that the authors had not verified those certificates and that the reported claims were not theorems. The bound in Theorem 1 is numerically smaller than both quantities. No historical priority or exact-value assertion is made.

2 A coefficient inequality for every finite matrix

Let γ be standard Gaussian measure on $\mathbb{R}$, with density $\phi(x) = (2\pi)^{-1/2}e^{-x^2/2}$. Define the probabilists' Hermite polynomials and their normalization by

$$e^{tx-t^2/2} = \sum_{k \geq 0} \text{He}_k(x) \frac{t^k}{k!}, \quad h_k = \frac{\text{He}_k}{\sqrt{k!}}.$$

Taking Gaussian expectations of two generating functions proves $\int h_j h_k d\gamma = \mathbf{1}_{j=k}$. These polynomials are complete in $L^2(\gamma)$: if u is orthogonal to all polynomials, then $\int u(x)e^{zx} d\gamma(x)$ is entire, by Cauchy–Schwarz uniformly on compact subsets of $\mathbb{C}$, and all its derivatives at zero vanish. Its values on the imaginary axis are the Fourier transform of the finite measure $u d\gamma$, so Fourier uniqueness gives $u = 0$. The same argument, or iteration in each variable, gives the tensor product basis in two dimensions. If (X, Y) are standard real Gaussians with covariance t , the generating functions give

$$\mathbb{E}[h_j(X)h_k(Y)] = \mathbf{1}_{j=k}t^j, \quad -1 \leq t \leq 1. \quad (2)$$

The endpoints follow either directly from $Y = \pm X$ or by continuity.

Let L be the following odd polynomial:

$$L(x) = \frac{3188129h_1(x) + 110767301h_3(x) + 13189406h_5(x) - 5120076h_7(x) - 1681617h_9(x)}{10^9}. \quad (3)$$

Set $f(w, x) = \text{sgn}(w + L(x))$ and $g(w, x) = \text{sgn}(w - L(x))$. The value of the sign at zero is immaterial, since conditioning on X shows that each threshold has Gaussian measure zero. Write

$$A_{jk} = \int_{\mathbb{R}^2} f(w, x)h_j(w)h_k(x) d\gamma(w) d\gamma(x).$$

Since $f^2 = 1$, Parseval gives $\sum_{j,k} A_{jk}^2 = 1$. Oddness under $(w, x) \mapsto (-w, -x)$ gives $A_{jk} = 0$ when $j + k$ is even. Moreover, $g(w, x) = f(w, -x)$, so its corresponding coefficient is $(-1)^k A_{jk}$.

For independent coordinate pairs (W, W') and (X, X') with covariances α and β , respectively, (2) gives

$$F(\alpha, \beta) := \frac{\pi}{2} \mathbb{E}[f(W, X)g(W', X')] = \frac{\pi}{2} \sum_{j,k \geq 0} (-1)^k A_{jk}^2 \alpha^j \beta^k. \quad (4)$$

This identity follows first by finite Hermite approximation and the L^2 contraction property of conditional expectation. The series is absolutely and uniformly convergent on the closed complex bidisk, since $\sum A_{jk}^2 = 1$. Thus it also defines a continuous function there, holomorphic in the open bidisk.

Let $P(t) = \sum_{d=1,3,\dots,81} p_d t^d$ be the exact rational polynomial in Appendix A. Its coefficient norm is

$$\rho_0 := \sum_d |p_d| = \frac{499995394004774337}{500000000000000000} < 1. \quad (5)$$

Define

$$H(t) = F(P(t), t) = \sum_{n \geq 0} b_n t^n. \quad (6)$$

The coefficients are real and vanish for even n . In the Banach algebra of absolutely summable power series, substitution in (4) gives

$$\sum_n |b_n| \leq \frac{\pi}{2} \sum_{j,k} A_{jk}^2 \|P\|_1^j \leq \frac{\pi}{2}. \quad (7)$$

This also justifies substitution and evaluation at $t = \pm 1$.

Lemma 2. *If the coefficients in (6) satisfy $\Gamma = b_1 - \sum_{n \geq 3, n \text{ odd}} |b_n| > 0$, then $V(M) \leq \pi S(M)/(2\Gamma)$ for every finite real matrix M .*

Proof. For real unit vectors u, v , form the direct sums

$$U_P(u) = \bigoplus_d \sqrt{|p_d|} u^{\otimes d} \oplus \sqrt{1 - \rho_0} e_L, \quad V_P(v) = \bigoplus_d \text{sgn}(p_d) \sqrt{|p_d|} v^{\otimes d} \oplus \sqrt{1 - \rho_0} e_R,$$

where e_L, e_R are orthogonal unit vectors in extra coordinates. Both vectors have norm one and their cross inner product is $P(\langle u, v \rangle)$. This is a finite-dimensional real construction; negative polynomial coefficients cause no loss of norm or positivity of the joint Gaussian covariance. Use a standard Gaussian vector in this space to obtain all W_i, W'_j simultaneously. In an independent Gaussian space use the original u_i, v_j to obtain X_i, X'_j . Each endpoint has independent standard W and X , and each cross pair has correlations $P(t_{ij})$ and t_{ij} , where $t_{ij} = \langle u_i, v_j \rangle$.

Choose vectors attaining $V(M)$, reversing all left vectors if needed so that $\sum M_{ij} t_{ij} = V(M)$. For every positive integer n , the unit vectors $u_i^{\otimes n}, v_j^{\otimes n}$ show that $|\sum M_{ij} t_{ij}^n| \leq V(M)$. Absolute summability in (7) and finiteness of M therefore imply

$$\mathbb{E} \sum_{i,j} M_{ij} f(W_i, X_i) g(W'_j, X'_j) = \frac{2}{\pi} \sum_{i,j} M_{ij} H(t_{ij}) \geq \frac{2}{\pi} \Gamma V(M).$$

For each Gaussian outcome the signed sum is at most $S(M)$. Taking expectations proves the assertion, also when $V(M) = 0$. $\square$

3 The complete coefficient calculation through degree 501

3.1 One-dimensional moments and exact finite algebra

Differentiating the Hermite generating function gives

$$h_{k+1}(x) = \frac{x h_k(x) - \sqrt{k} h_{k-1}(x)}{\sqrt{k+1}}, \quad h'_k(x) = \sqrt{k} h_{k-1}(x). \quad (8)$$

For $a \geq 0$, put $q_a(u) = \int \operatorname{sgn}(w+u) h_a(w) d\gamma(w)$. The identity $(h_{a-1}\phi)' = -\sqrt{a} h_a \phi$, including the vanishing boundary term at infinity, gives

$$\begin{aligned} q_0(u) &= \operatorname{erf}(u/\sqrt{2}), \\ q_a(u) &= \sqrt{\frac{2}{\pi}} e^{-u^2/2} \frac{h_{a-1}(-u)}{\sqrt{a}} \quad (a \geq 1), \\ q_{a+1}(u) &= \frac{-u\sqrt{a} q_a(u) - (a-1)q_{a-1}(u)}{\sqrt{a(a+1)}} \quad (a \geq 1). \end{aligned} \tag{9}$$

Consequently

$$A_{ab} = \int_{\mathbb{R}} h_b(x) q_a(L(x)) d\gamma(x). \tag{10}$$

For real u , Cauchy–Schwarz gives $|q_a(u)| \leq 1$ for every a . Thus the entire omitted real Gaussian tail satisfies

$$\left| \int_{|x|>16} h_b(x) q_a(L(x)) d\gamma(x) \right| \leq \sqrt{\operatorname{erfc}(16/\sqrt{2})}. \tag{11}$$

For $a+b$ odd the integrand in (10) is even.

We calculate every moment in

$$\mathcal{I} = \{(a, b) \in \mathbb{Z}_{\geq 0}^2 : a+b \leq 501, a+b \text{ odd}\}.$$

There are $\sum_{r=0}^{250} (2r+2) = 63252$ such moments. Since $P(0) = 0$, $P(t)^a t^b$ has order at least $a+b$. It follows rigorously that no moment outside $\mathcal{I}$ contributes to any coefficient of degree at most 501. In particular,

$$b_n = \frac{\pi}{2} \sum_{\substack{a,b \geq 0 \\ a+b \leq n}} (-1)^b A_{ab}^2 [t^{n-b}] P(t)^a \quad (1 \leq n \leq 501). \tag{12}$$

All powers of P may be computed in $\mathbb{Q}[t]/(t^{502})$; reduction modulo t^{502} is a ring homomorphism. There are 251 odd coefficients $b_1, b_3, \dots, b_{501}$, including b_1 .

3.2 Nodes, weights, and a uniform quadrature remainder

We use 128 equal panels on $[0, 16]$, with half-width $h = 1/16$, and 112-point Gauss–Legendre quadrature on each panel. To specify why the quadrature is rigorous, let $\mathcal{P}_s$ denote the Legendre polynomial with $\mathcal{P}_s(1) = 1$. Rodrigues’ formula, followed by integration by parts, gives orthogonality to polynomials of degree less than s and $\int_{-1}^1 \mathcal{P}_s^2 = 2/(2s+1)$. There are 112 disjoint rational brackets inside $(-1, 1)$, with exactly checked opposite signs of $\mathcal{P}_{112}$ at their endpoints. Degree 112 therefore implies that they contain all its roots, each simple. For a root z_i , the Lagrange polynomial $\ell_i(x) = \mathcal{P}_{112}(x)/((x-z_i)\mathcal{P}'_{112}(z_i))$ gives weights $w_i = \int_{-1}^1 \ell_i^2 > 0$. Polynomial division by $\mathcal{P}_{112}$ shows that $\ell_i^2 - \ell_i$ has integral zero, so $w_i = \int_{-1}^1 \ell_i$. For any polynomial of degree at most 223, division gives a multiple of $\mathcal{P}_{112}$ with quotient of degree at most 111, plus an interpolating remainder of degree at most 111. Orthogonality and Lagrange interpolation therefore prove exactness through degree 223. The Christoffel–Darboux identity, obtained by telescoping the three-term Legendre recurrence, gives

$$w_i = \frac{2}{(1-z_i^2)\mathcal{P}'_{112}(z_i)^2}, \quad \sum_i w_i = 2. \tag{13}$$

For completeness, the diagonal identity used here is

$$\sum_{k=0}^{s-1} \frac{2k+1}{2} \mathcal{P}_k(x) \mathcal{P}_k(y) = \frac{s}{2} \frac{\mathcal{P}_s(x) \mathcal{P}_{s-1}(y) - \mathcal{P}_{s-1}(x) \mathcal{P}_s(y)}{x-y};$$

at a root of $\mathcal{P}_s$, use $(1 - z_i^2) \mathcal{P}'_s(z_i) = s \mathcal{P}_{s-1}(z_i)$.

All root and weight evaluations use enclosing intervals. A midpoint derivative enclosure is enlarged by $\sup_{[-1,1]} |\mathcal{P}_{112}''|$ times the bracket radius. The exact bound is

$$\sup_{[-1,1]} |\mathcal{P}_s''| \leq \frac{(s-1)s(s+1)(s+2)}{8}.$$

Indeed $\mathcal{P}_s'' = 3C_{s-2}^{5/2}$; the generating function $(1 - 2xt + t^2)^{-5/2}$ and its factorization for $x = \cos \theta$ bound the absolute value of every coefficient by its value at $x = 1$. Exact rational sign checks, disjointness checks, and further rational bisections independently enclose all nodes and the weights (13). Initial floating-point guesses play no evidentiary role.

We next bound the integrand on full complex ellipses, uniformly for all moments. For $0 < \tau < 1$ and $z = x + iy$, Mehler's identity implies

$$\sum_{k \geq 0} \tau^k |h_k(z)|^2 = (1 - \tau^2)^{-1/2} \exp \left(\frac{\tau x^2}{1 + \tau} + \frac{\tau y^2}{1 - \tau} \right). \quad (14)$$

One direct justification, valid for complex z , uses the entire kernel

$$K_s(z, u) = \frac{1}{\sqrt{1 - s^2}} \exp \left(\frac{2szu - s^2(z^2 + u^2)}{2(1 - s^2)} \right), \quad s = \sqrt{\tau}.$$

Gaussian integration of its product with the Hermite generating function shows that its Hermite coefficients in u are $s^k h_k(z)$. It belongs to $L^2(\gamma(du))$ for each complex z . Completing the square in $\int |K_s(z, u)|^2 d\gamma(u)$ and using Parseval gives (14), including its convergence assertion.

Fix $N = 501$, $\tau = 99/100$, and $\rho = 3/2$, and let $F_N = \tau^{-N/2} (1 - \tau^2)^{-1/4}$. Dropping a term from (14) gives, for $b \leq N$,

$$|h_b(z) \phi(z)| \leq \frac{F_N}{\sqrt{2\pi}} \exp \left(-\frac{(\Re z)^2}{2(1 + \tau)} + \frac{(\Im z)^2}{2(1 - \tau)} \right). \quad (15)$$

Equation (9) gives the same exponential bound for $q_a(u)$, with prefactor $\sqrt{2/\pi} F_N$, when $1 \leq a \leq N$. For q_0 , integrate its derivative along a vertical segment to obtain

$$|q_0(u)| \leq 1 + \sqrt{2/\pi} |\Im u| \exp((\Im u)^2 - (\Re u)^2)/2. \quad (16)$$

For a panel centered at $c = (2j + 1)h$, define

$$x_r = h(\rho + \rho^{-1})/2, \quad y_r = h(\rho - \rho^{-1})/2, \quad R = (x_r^2 + y_r^2)^{1/2}.$$

The closed Bernstein ellipse for this panel lies in $|\Re z - c| \leq x_r$, $|\Im z| \leq y_r$, $|z - c| \leq R$. Exact polynomial Taylor expansion bounds $|\Re L(z)|$ below by

$$U_c = \max \left\{ 0, |L(c)| - \sum_{k=1}^9 \frac{|L^{(k)}(c)| R^k}{k!} \right\},$$

and bounds $|\Im L(z)|$ above by

$$V_c = y_r \sum_{k=0}^8 \frac{|L^{(k+1)}(c)| R^k}{k!}.$$

The latter bound follows by integrating L' along the vertical segment from $\Re z$ to z ; L is real on the real axis. Also $|\Re z| \geq X_c := \max\{0, c - x_r\}$. Thus a uniform upper bound on $|h_b(z)q_a(L(z))\phi(z)|$ on the entire ellipse is

$$M_c = \frac{F_N}{\sqrt{2\pi}} e^{-X_c^2/(2(1+\tau)) + y_r^2/(2(1-\tau))} \cdot \max \left\{ \sqrt{2/\pi} F_N e^{-U_c^2/(2(1+\tau)) + V_c^2/(2(1-\tau))}, 1 + \sqrt{2/\pi} V_c e^{(V_c^2 - U_c^2)/2} \right\}. \quad (17)$$

If an analytic function has absolute value at most M on a Bernstein ellipse of parameter ρ , its Chebyshev coefficients of positive degree have absolute value at most $2M\rho^{-k}$, by the Cauchy formula after $z = (w + w^{-1})/2$. Truncation after degree 223 therefore has uniform error at most $2M\rho^{-224}/(1-\rho^{-1})$ on $[-1, 1]$. Both integration and the positive Gauss rule have functional norm two on that interval. Exactness through degree 223 now bounds their difference by $8M\rho^{-224}/(1-\rho^{-1})$. Scaling each panel and doubling the half-line integral gives the uniform complete error

$$E_A = \frac{16h\rho^{-224}}{1-\rho^{-1}} \sum_{j=0}^{127} M_{(2j+1)h} + \sqrt{\operatorname{erfc}(16/\sqrt{2})} < 4.121 \cdot 10^{-22}. \quad (18)$$

The terms are evaluated with outward rounding, including all square roots and transcendental functions. The first term is less than $4.1200500 \cdot 10^{-22}$ and the second less than $3.574565 \cdot 10^{-29}$. This bound uses $N = 501$ everywhere in F_N ; it is not the degree-251 quadrature error.

3.3 Finite certificate and resulting coefficient inequalities

The accompanying certificate contains the 112 node enclosures, 128 ellipse bounds, and 32 blocks of four panels each. Each block contains every index in $\mathcal{I}$. The blocks are summed with interval arithmetic, after which the symmetric error interval $[-E_A, E_A]$ is added once to each moment. No panel is omitted or repeated. The moment calculation uses 384-bit working precision, with 640 bits for initial node refinement. All decimal intervals are serialized as $(m \pm r)10^e$ with integer m, r, e and are read with outward rounding.

Formula (12) is then evaluated in two ways: by interval power-series arithmetic and by exact rational powers of P followed by interval evaluation of the moment factors. Both contain every odd coefficient through 501 with its factor $(-1)^b$ unchanged. The exact rational implementation gives the strict outward inequalities

$$b_1 > B := 0.88182999999999823752, \quad \sum_{\substack{3 \leq n \leq 501 \\ n \text{ odd}}} |b_n| < h_* := 0.000000001054. \quad (19)$$

For orientation, the interval sum before this rational enlargement is $[1.0537295443 \cdot 10^{-9} \pm 5.65 \cdot 10^{-20}]$. Only the outward inequalities (19) are used below.

4 Regularity and a bound on the entire unit circle

Let $D = z d/dz$. Since $d/d\theta = iD$ on a circle, the fourth angular derivative is exactly D^4 . We will prove

$$\left(\frac{1}{2\pi} \int_0^{2\pi} |D^4 H(e^{i\theta})|^2 d\theta \right)^{1/2} < C_* := 24744.587741. \quad (20)$$

Two different estimates for the mixed derivatives of F are used on closed angular intervals. Before giving the numerical estimates, we establish the boundary regularity needed to interpret this integral and to pass from it to coefficient bounds.

4.1 Gaussian smoothing in one coordinate

For $0 < r < 1$, put $v = 1 - r$ and

$$G_r(w, x) = \mathbb{E}_Z f(\sqrt{r}w + \sqrt{1-r}Z, x) = \operatorname{erf}\left(\frac{\sqrt{r}w + L(x)}{\sqrt{2v}}\right).$$

Here Z is an independent standard real Gaussian. The generating function or (2) shows that G_r has Hermite coefficient $r^{j/2}A_{jk}$. For nonnegative p, q with $p + q \geq 1$ define

$$S_{pq}(r) = r^{-p} \|\partial_w^p \partial_x^q G_r\|_{L^2(\gamma^2)}^2 = \sum_{j,k} (j)_p (k)_q r^{j-p} A_{jk}^2, \quad (21)$$

where $(j)_p = j!/(j-p)!$ if $j \geq p$ and is zero otherwise. To justify this identity without assuming differentiability of f , note that every fixed derivative of the explicit smooth function G_r is a polynomial in w, x times a Gaussian in $\sqrt{r}w + L(x)$. It and all lower derivatives have at most polynomial growth and belong to $L^2(\gamma^2)$. Gaussian integration by parts therefore has vanishing boundary terms. It identifies the Hermite coefficients of the derivative as the shifted coefficients of G_r multiplied by $\sqrt{(j)_p (k)_q}$. Completeness and Parseval prove (21), including finiteness.

Termwise differentiation of (4) now gives

$$|\partial_\alpha^p \partial_\beta^q F(\alpha, \beta)| \leq \frac{\pi}{2} S_{pq}(r), \quad |\alpha| \leq r, \quad |\beta| \leq 1. \quad (22)$$

The differentiated series converges absolutely and uniformly on this set, by (21); terms with insufficient indices vanish. Taking $r = (1 + \rho_0)/2$ and using $|P(z)| \leq \rho_0$ on $|z| \leq 1$ proves that every mixed derivative needed through order four is continuous after composition on the closed disk. The chain rule consequently gives a continuous boundary value for $D^4 H$, with uniform convergence of $D^4 H(se^{i\theta})$ as $s \uparrow 1$. This includes $z = 1$ and $z = -1$. This qualitative argument requires no numerical evaluation at the parameter $(1 + \rho_0)/2$.

We next give an explicit finite calculation of $S_{pq}(r)$ for $1 \leq p + q \leq 4$. Put $T = \sqrt{r}w + L(x)$ and $\psi(T) = \operatorname{erf}(T/\sqrt{2v})$. For $d \geq 1$,

$$\psi^{(d)}(T) = \sqrt{2/\pi} v^{-1/2} e^{-T^2/(2v)} U_d(T), \quad U_d(T) = (-1)^{d-1} v^{-(d-1)/2} \operatorname{He}_{d-1}(T/\sqrt{v}).$$

In particular,

$$U_1 = 1, \quad U_2 = -T/v, \quad U_3 = T^2/v^2 - 1/v, \quad U_4 = -T^3/v^3 + 3T/v^2.$$

The chain rule gives

$$\partial_w^p \partial_x^q G_r = r^{p/2} \sqrt{2/\pi} v^{-1/2} e^{-T^2/(2v)} R_{pq}(x, T), \quad R_{pq} = \sum_k B_{qk} U_{p+k}. \quad (23)$$

Here $B_{00} = 1$; for $q > 0$ the nonzero Bell polynomials are

$$\begin{aligned} B_{11} &= L', \\ B_{21} &= L'', & B_{22} &= (L')^2, \\ B_{31} &= L''', & B_{32} &= 3L'L'', & B_{33} &= (L')^3, \\ B_{41} &= L'''' , & B_{42} &= 4L'L''' + 3(L'')^2, & B_{43} &= 6(L')^2L'', & B_{44} &= (L')^4. \end{aligned}$$

These identities follow by differentiating the previous row and collecting derivatives of ψ ; no asymptotic chain rule is involved. Only $p + k \geq 1$ occurs when $p + q \geq 1$.

For any polynomial Q_0 and fixed real x , completing the square in the w integral proves

$$\mathbb{E}_w[e^{-T^2/v} Q_0(T)] = \sqrt{\frac{v}{1+r}} e^{-L(x)^2/(1+r)} \mathbb{E}[Q_0(Y)], \quad Y \sim N\left(\frac{vL(x)}{1+r}, \frac{rv}{1+r}\right). \quad (24)$$

Indeed the original density has variance r and mean $L(x)$ in the T coordinate; adding T^2/v to its exponent gives the stated mean, variance, and normalizing factor. The moments of Y , with mean μ and variance σ^2 , are exactly

$$\mathbb{E}Y^d = \sum_{\ell=0}^{\lfloor d/2 \rfloor} \frac{d!}{2^\ell \ell! (d-2\ell)!} \sigma^{2\ell} \mu^{d-2\ell}, \quad (25)$$

as follows by expanding $\exp(\mu t + \sigma^2 t^2/2)$. With the real polynomial $Q_{pq}(x) = \mathbb{E}[R_{pq}(x, Y)^2]$, equations (21)–(25) give

$$S_{pq}(r) = \frac{2}{\pi \sqrt{1-r^2}} \int_{\mathbb{R}} Q_{pq}(x) e^{-L(x)^2/(1+r)} d\gamma(x). \quad (26)$$

This polynomial is even, is nonnegative on the real line, and has degree at most 118 in the cases used here. Its coefficients need not be nonnegative. Evenness also follows directly from simultaneous sign change of w, x in the squared derivative in (23).

There is a second way to construct exactly the same polynomial, used in the accompanying implementation. For polynomials in T with polynomial coefficients in x , let

$$\mathcal{T}R = \partial_T R - TR/v, \quad \mathcal{X}R = \partial_x R + L'(x)\mathcal{T}R.$$

If $p > 0$, start from 1, apply $\mathcal{T}$ a total of $p - 1$ times, then $\mathcal{X}$ a total of q times. If $p = 0$, start from L' and apply $\mathcal{X}$ a total of $q - 1$ times. Differentiating $e^{-T^2/(2v)}R$ verifies these rules directly. Squaring the result and using (25) gives Q_{pq} , providing a finite construction independent of the Bell polynomial listing.

4.2 The complete error for the smoothing integrals

Every finite integral in (26) is evaluated on $[0, 16]$ as the sum of the 16 integrals on $[j, j + 1]$, then doubled. The complex integrand supplied to the interval integrator is the entire function

$$J(z) = \frac{2Q_{pq}(z)}{\pi \sqrt{1-r^2} \sqrt{2\pi}} \exp\left(-\frac{L(z)^2}{1+r} - \frac{z^2}{2}\right). \quad (27)$$

In particular, the square in this formula is the analytic square, not a complex absolute square. Enclosing polynomial coefficients enclose the exact algebraic coefficients of L and Q_{pq} .

The finite-interval calculation uses the certified complex integration algorithm in Arb/FLINT [5, 6, 7]. Its adaptive Gauss–Legendre rule encloses the integrand on a full complex ellipse; it accepts a quadrature result only with a rigorous remainder. For an ellipse of parameter $\rho > 1$, maximum bound M , and s nodes, the bound used by that algorithm on $[-1, 1]$ is

$$\frac{64M}{15(\rho - 1)\rho^{2s-1}}. \quad (28)$$

It is the analytic Gauss–Legendre remainder bound implemented in the cited algorithm; scaling by the interval half-width gives the bound on a subinterval. A fully elementary, more conservative remainder $8M\rho^{-2s}/(1-\rho^{-1})$ follows from the Chebyshev proof above and also explains why such enclosing quadratures converge here. Subdivision adds enclosing integrals and their errors, rather than extrapolating point samples. Direct integration enclosures on small intervals are also permitted by the algorithm. All calls use 384-bit arithmetic, relative tolerance 10^{-18} , absolute tolerance 10^{-20} , and an evaluation limit of 10^6 ; these tolerances request accuracy and are not themselves claimed as error bounds. Only the returned finite enclosures are used. Entire analyticity of (27) validates every analyticity query and every ellipse bound required by the integrator. The mathematical dependence here is the proved enclosing quadrature algorithm and the inclusion property of ball arithmetic, not an uncertified numerical integration routine.

The infinite remainder is bounded separately. Write $L(x) = \sum_{k=0}^9 \lambda_k x^k$. The coefficient enclosures prove

$$\sum_{k<9} |\lambda_k| 16^{k-9} < \frac{|\lambda_9|}{2}, \quad c := 10^{-6} < \frac{|\lambda_9|}{2}. \quad (29)$$

Hence $|L(x)| \geq c|x|^9$ for $|x| \geq 16$. If $m = \deg Q_{pq}$, the further checked inequality

$$\frac{18c^2 16^{18}}{1+r} > m$$

shows that $x^k e^{-c^2 x^{18}/(1+r)}$ decreases for $x \geq 16$ and every $0 \leq k \leq m$. Therefore the entire two-sided omitted integral in (26) is at most

$$\frac{2}{\pi\sqrt{1-r^2}} \left(\sum_{k=0}^m |[x^k]Q_{pq}| 16^k \right) e^{-c^2 16^{18}/(1+r)} < 2^{-100}. \quad (30)$$

In deriving this bound we used only that the Gaussian measure of $\{|x| > 16\}$ is at most one, so both tails are included. A symmetric interval of this radius is added to twice the finite real integral. All 266 required integral records contain the finite integral enclosure, this tail bound, and the positive margins in (29) and the monotonicity test. These records are computed from an empty integral cache.

4.3 Mixed derivative bounds from complex Gaussian densities

For $z \in \mathbb{C}$ with $|\Re z| < 1$, the analytic continuation of a correlated Gaussian density, after the real orthogonal change of variables $U = (x+y)/\sqrt{2}$, $V = (x-y)/\sqrt{2}$, is

$$k_z(U, V) = \frac{1}{2\pi\sqrt{1+z}\sqrt{1-z}} \exp\left(-\frac{U^2}{2(1+z)} - \frac{V^2}{2(1-z)}\right).$$

Each square root is its analytic branch in the right half-plane. On compact subsets of the strip the real parts of the inverse variances are positive and uniformly bounded below. Thus this density and every fixed parameter derivative have an integrable Gaussian majorant times a polynomial. Differentiation of its integral against bounded real functions is justified by dominated convergence and defines a holomorphic function.

Let $d_0(z) = \int_{\mathbb{R}^2} |k_z|$. Completing two real Gaussian integrals gives

$$d_0(z) = \sqrt{\frac{|1+z||1-z|}{1-(\Re z)^2}}, \quad v_U = \frac{|1+z|^2}{1+\Re z}, \quad v_V = \frac{|1-z|^2}{1-\Re z}. \quad (31)$$

The probability density $|k_z|/d_0(z)$ is that of independent centered real Gaussians with variances v_U, v_V . To compute derivatives write $\partial_z^p k_z = k_z T_p(U, V; z)$. One explicit formula is

$$\begin{aligned} R_j(v, u) &= \sum_{\ell=0}^j \frac{(2j)!(-1)^{j-\ell}}{2^{2j-\ell}(j-\ell)!(2\ell)!} v^{-j-\ell} u^{2\ell}, \\ T_p &= \sum_{j=0}^p \binom{p}{j} (-1)^{p-j} R_j(1+z, U) R_{p-j}(1-z, V). \end{aligned} \quad (32)$$

For a one-dimensional Gaussian density, the heat equation $\partial_v = \frac{1}{2}\partial_u^2$ and the Hermite formula for its spatial derivatives prove R_j . The product rule and $\partial_z(1-z) = -1$ then prove (32). Cauchy-Schwarz under the probability law just described gives

$$\|\partial_z^p k_z\|_{L^1(\mathbb{R}^2)} \leq d_p(z) := d_0(z) \sqrt{\mathbb{E}|T_p(U, V; z)|^2}. \quad (33)$$

All expectations here are finite polynomial Gaussian moments, without any integral truncation.

For an alternative exact construction, the k th logarithmic derivative $\ell_k = \partial_z^k \log k_z$ has constant, U^2 , and V^2 coefficients respectively

$$\begin{aligned} &\frac{(-1)^k(k-1)!}{2(1+z)^k} + \frac{(k-1)!}{2(1-z)^k}, \\ &\frac{(-1)^{k+1}k!}{2(1+z)^{k+1}}, \quad -\frac{k!}{2(1-z)^{k+1}}. \end{aligned}$$

Starting with $T_0 = 1$, the finite recurrence

$$T_m = \sum_{k=1}^m \binom{m-1}{k-1} \ell_k T_{m-k}$$

is obtained by differentiating an exponential generating function for k_{z+t}/k_z . It is the construction used in the independent implementation. To evaluate its squared norm without a cancellation assumption, rescale $U = \sqrt{v_U} u$, $V = \sqrt{v_V} v$, and use

$$u^{2k} = \sum_{i=0}^k \frac{(2k)!}{2^{k-i}(k-i)!(2i)!} \text{He}_{2i}(u).$$

This identity follows by comparing coefficients in the Hermite generating function multiplied by $e^{t^2/2}$. If the resulting polynomial is $\sum_{i,j} c_{ij} \text{He}_{2i}(u) \text{He}_{2j}(v)$, then

$$\mathbb{E}|T_p|^2 = \sum_{i,j} |c_{ij}|^2 (2i)!(2j)!.$$

Thus all complex conjugations occur only in a finite nonnegative square norm, not in the analytic density or its derivative.

Using independent copies of the correlated density for the W and X coordinates, boundedness $|fg| \leq 1$ gives

$$|\partial_\alpha^p \partial_\beta^q F(\alpha, \beta)| \leq \frac{\pi}{2} d_p(\alpha) d_q(\beta), \quad |\Re \alpha|, |\Re \beta| < 1. \quad (34)$$

To identify this analytic function with (4), first note their equality on the real open square of correlations. Fixing a real second coordinate and using the one-variable identity theorem, then fixing a complex first coordinate and repeating, proves equality in the open bidisk. At boundary points within the strip, the derivatives agree with the continuous limits established by (21). This proves that (34) bounds the same derivatives used by the coefficient argument.

4.4 Angular chain rule and complete interval coverage

For $s \geq 0$ write

$$D^s H(z) = \sum_{p,q} c_{pq}^{(s)}(z) (\partial_\alpha^p \partial_\beta^q F)(P(z), z).$$

These are exact rational polynomial coefficients. For example they are given by the finite formula

$$c_{pq}^{(s)}(z) = \frac{s!}{p!q!} [u^s] (P(ze^u) - P(z))^p (ze^u - z)^q. \quad (35)$$

Taylor's formula for $F(P(ze^u), ze^u)$ proves this identity in the open disk. It can also be computed recursively, with out-of-range indices zero, from

$$c_{00}^{(0)} = 1, \quad c_{pq}^{(s+1)} = Dc_{pq}^{(s)} + (DP)c_{p-1,q}^{(s)} + zc_{p,q-1}^{(s)}. \quad (36)$$

For $s = 4$ the nonzero indices are precisely $p, q \geq 0$, $1 \leq p + q \leq 4$, a set of 14 elements. All these polynomials are computed over $\mathbb{Q}$, so the finite chain rule contains no numerical differentiation error.

Divide the first quadrant into the 512 closed intervals

$$I_j = [j\pi/1024, (j+1)\pi/1024], \quad 0 \leq j < 512.$$

Interval sine and cosine of kI_j enclose $z^k = e^{ik\theta}$ for every $\theta \in I_j$. This yields enclosures for $P(z)$ and all $c_{pq}^{(4)}(z)$ on each whole interval. A smoothing bound is available whenever the enclosure proves $|P(z)| < r$ for one of $r = 78/100, 79/100, \dots, 97/100$. A density bound is available when both $|\Re P(z)| < 1$ and $|\Im z| < 1$ are proved by the same interval enclosures. Every panel has at least one available estimate. Where both are available, their smaller upper bound may be used separately for each (p, q) . The intervals touching $z = 1$ are covered by smoothing; the density estimate is not asserted at its singular endpoint.

Let $U_{j,pq}$ be such an upper bound for $(2/\pi)|\partial_\alpha^p \partial_\beta^q F|$, and let $C_{j,pq}$ bound $|c_{pq}^{(4)}|$. Then

$$|D^4 H(e^{i\theta})| \leq M_j := \frac{\pi}{2} \sum_{1 \leq p+q \leq 4} C_{j,pq} U_{j,pq} \quad (\theta \in I_j). \quad (37)$$

The circle certificate records all $512 \cdot 14 = 7168$ terms, each chosen method, and its domain inequalities. Exactly 266 smoothing integrals are required: all 14 indices at each of the 19 parameters $79/100, \dots, 97/100$. These are the integrals bounded in (26)–(30).

Real coefficients and oddness imply $D^4 H(\bar{z}) = \overline{D^4 H(z)}$ and $D^4 H(-z) = -D^4 H(z)$. The four quadrants therefore give equal contributions to its squared norm. Round each M_j upwards to $\widehat{M}_j = k_j/10^8$ with integer k_j . The complete list of these 512 rational upper bounds is included in the certificate. Summing their exact rational squares gives

$$\frac{1}{2\pi} \int_0^{2\pi} |D^4 H(e^{i\theta})|^2 d\theta \leq \frac{1}{512} \sum_{j=0}^{511} \widehat{M}_j^2 = Q := \frac{1567474233421402382277578179}{2560000000000000000}. \quad (38)$$

The explicit strict rational comparison

$$C_*^2 - Q = \frac{107039174409781821}{2560000000000000000} > 0 \quad (39)$$

proves (20). The bounds concern complete angular intervals, not samples of the circle.

5 The coefficient tail and the exact numerical conclusion

For $0 < s < 1$, ordinary Fourier Parseval applied to the power series gives

$$\frac{1}{2\pi} \int_0^{2\pi} |D^4 H(se^{i\theta})|^2 d\theta = \sum_{n \geq 0} n^8 |b_n|^2 s^{2n}.$$

The continuous boundary extension proved above permits passage to the limit on the left by uniform convergence. Monotone convergence of the nonnegative series gives the limit on the right. Hence

$$\sum_{n \geq 0} n^8 |b_n|^2 = \frac{1}{2\pi} \int_0^{2\pi} |D^4 H(e^{i\theta})|^2 d\theta < C_*^2. \quad (40)$$

For every positive odd N , Cauchy–Schwarz and the decreasing function $(N + 2x)^{-8}$ show that

$$\begin{aligned} \sum_{\substack{n>N \\ n \text{ odd}}} |b_n| &\leq \left(\sum_{n>N} n^8 |b_n|^2 \right)^{1/2} \left(\sum_{k=1}^{\infty} (N+2k)^{-8} \right)^{1/2} \\ &< \frac{C_*}{\sqrt{14N^7}}, \end{aligned} \quad (41)$$

because the latter sum is at most $\int_0^\infty (N+2x)^{-8} dx = 1/(14N^7)$. For $N = 501$ the first omitted coefficient is b_{503} . Put

$$T_* = 0.000002349547147305, \quad \gamma_* = 0.8818276493988.$$

All these displayed terminating decimals denote exact rational numbers. The certificate checks the following integer comparison:

$$T_*^2(14 \cdot 501^7) - C_*^2 = \frac{3104661541167915364988902768147}{2000000000000000000000000000000} > 0. \quad (42)$$

Thus (19) and (41) give

$$\begin{aligned} \Gamma &> B - h_* - T_* \\ &= 0.88182764939885093252 = \frac{22045691234971273313}{25000000000000000000} > \gamma_*, \\ B - h_* - T_* - \gamma_* &= \frac{1273313}{25000000000000000000} > 0. \end{aligned} \quad (43)$$

In particular, the coefficient margin is positive with a fixed uniform constant, independent of the dimensions or entries of M .

We give an elementary rational verification of the last reciprocal comparison, so it does not depend on a decimal approximation to π . Machin's identity is

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239).$$

It follows, for example, by applying the tangent addition law to $4 \arctan(1/5) - \arctan(1/239)$ and observing that this angle lies in $(0, \pi/2)$ and has tangent one. For $0 < x < 1$, let $s_{64}(x) = \sum_{k=0}^{63} (-1)^k x^{2k+1} / (2k+1)$. The alternating series remainder gives

$$s_{64}(x) < \arctan x < s_{64}(x) + x^{129}/129.$$

1. The exact input is `head/sources/candidate_rational.json`. Its polynomial coefficients are printed in (3) and Appendix A. Its SHA-256 hash is

207b7ed03d6bc1fd44922863e361544bb87442d13801870f0a8bc6705e8a3040.

Only the exact rational coefficient strings determine the polynomials; diagnostic fields in the file are not assumptions.

2. The moment calculation uses `head/run/gauss_nodes_112.json`, `head/run/head501_config.json`, and all 32 files in `head/run/blocks/`. The merged file is `head/run/head_moments_501.json`. Its full index set is $\mathcal{I}$; the complete quadrature and infinite remainders are (18) and (11). The configuration binds the input, source, node, and block parameters. Each block is bound to that configuration and contains exactly its prescribed four panels.
3. The two complete composed coefficient files are `head/run/forward_head_501.json` and `head/evidence/independent_head.json`. Exact rational powers and the index proof in (12) establish that no coefficient through degree 501 is missing. Moment arithmetic and the rational-power assembly use 384 bits; the separate interval-series assembly uses 160 bits. The exact node containment calculation uses rational arithmetic and does not accept approximate root locations.
4. The norm calculation uses the files `independent_integrals.json` and `independent_circle.json` in `norm/evidence/`. These contain all 266 integrals and all 512 closed panels with their 14 terms, including the domain inequalities for each estimate. Equations (26)–(30) justify the integral values and their complete errors; equations (31)–(37) justify every other finite term. The filename and the auxiliary degree-251 tail field of the norm engine do not limit its derivative bound: its integral and circle calculations depend only on L , P , and derivative order four. The degree-501 tail is calculated separately from (41).
5. The file `rational_certificate.json` contains every integer k_j in (38), the exact rational head endpoints, and all the rational inequalities (39) and (42)–(44). It binds the coefficient, integral, and circle files by their hashes. The rational caps are at least both the saved panel majorants and independently re-summed products of all their saved term enclosures.

The arithmetic implementation uses Python, python-flint 0.8.0, and FLINT 3.3.1. Its mathematical requirements are inclusion-preserving real and complex ball operations, correct exact rational arithmetic, and the certified integral algorithm described above. Floating-point Legendre initializers, approximate optimization data, stored success flags, and prior integral caches are not premises. SciPy is optional: if unavailable, stored node midpoints provide initial guesses, after which all brackets, signs, and weights are proved again.

From the directory containing `verify.py`, the command

```
python -B -u verify.py -output /tmp/real-upper-verification
```

checks hashes and numerical bindings, reproduces the complete moment merge and the uniform error, verifies the nodes, reruns both coefficient assemblies, and recomputes all smoothing integrals and circle panels from an empty integral cache. It then checks the exact rational conclusion. Recomputed integral balls may have different endpoints because of adaptive subdivision or arithmetic versions; each must be a valid enclosure and must itself give $C_4 < C_*$. In particular, the printed value of Q specifies the supplied rational certificate and does not require bitwise equality of independently recomputed quadrature results.

The separate program `reproduce.py` initializes new nodes and quadrature errors and recomputes every moment block, with one file per block so that interruptions can be resumed. It then performs both assemblies, a new integral and circle calculation, and the rational comparison. It neither installs software nor acquires computational resources. Its command-line options and complete reproduction commands are in `README.txt` and `commands.txt`. The proof of the infinite coefficient bound is the regularity and Parseval argument in the main text; no finite file or numerical sample substitutes for that argument.

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